

Auteur: Thijs Van Schel
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$$f(x, y) = x^4 + y^4 - 4xy$$

$$\begin{cases} \frac{\partial f}{\partial x} = 4x^3 - 4y = 0 \\ \frac{\partial f}{\partial y} = 4y^3 - 4x = 0 \end{cases} \Rightarrow \begin{cases} y = x^3 \\ (x^3)^3 - x = 0 \end{cases} \Rightarrow \begin{cases} y = x^3 \\ x(x^8 - 1) = 0 \end{cases} \Rightarrow \begin{cases} x \in \{-1, 0, 1\} \\ y \in \{-1, 0, 1\} \end{cases}$$
$$\Rightarrow (x, y) \in \{(-1, -1), (0, 0), (1, 1)\}$$

$$\begin{cases} \frac{\partial^2 f}{\partial x^2} = 12x^2 \\ \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = -4 \\ \frac{\partial^2 f}{\partial y^2} = 12y^2 \end{cases} \Rightarrow H(x, y) = 144x^2y^2 - 16$$

- $H(-1, -1) = 128 > 0$ en $\frac{\partial^2 f}{\partial x^2}(-1, -1) = 12 > 0 \Rightarrow (-1, -1)$ minimum
- $H(0, 0) = -16 < 0 \Rightarrow (0, 0)$ zadelpunt
- $H(1, 1) = 128 > 0$ en $\frac{\partial^2 f}{\partial x^2}(1, 1) = 12 > 0 \Rightarrow (1, 1)$ minimum

References